Image Sharpening

- Image sharpening
  - Spatial Domain
    - Firstorder
    - Secondorder
  - Frequency Domain
    - Lowpass
    - Highpass

First order
* Robert's cross
  Prewitt operator
  Sopell operator

* Laplacian operation

Gradiant magnitude :- The strength of the edge is measured using gradiant magnitude.

* It is calculated using horizontal, vertical components.

$$G = \sqrt{G_x^2 + G_y^2}$$

G_x - horizontal gradient
G_y - vertical gradient

approximate Gradient magnitude

$$G = |G_x| + |G_y|$$

Gradian Direction

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

Robert's Cross operator :-

* Here mass or kernel is 2×2.

$G_x = f(x,y) - f(x+1, y+1)$

$G_y = f(x+1, y) - f(x, y+1)$.

$G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

15	20	30	35
25	35	50	55
40	60	75	85
50	70	95	110

Find the G_x value

$G_x \begin{bmatrix} 15 & 20 \\ 25 & 35 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$= 15(1) + 20(0) + 25(0) + (0)\ 35(-1)$

$= 15 - 35$

$= -20$

$G_y = \begin{bmatrix} 15 & 20 \\ 25 & 35 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 15(0) + 20(+1) + 25(-1) + 35(0)$

$= -5$

$G = \sqrt{G_x^2 + G_y^2}$

$G = \sqrt{400 + 25}$

$G = 20.61$

$G_x \begin{bmatrix} 20 & 30 \\ 35 & 50 \end{bmatrix} \begin{bmatrix} 01 & 0 \\ 0 & -1 \end{bmatrix}$

$= 20 + 0 + 0 - 50$

$G_x = -30$

$G_y = \begin{bmatrix} 20 & 30 \\ 35 & 50 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$= 0 + 30 + (-35) + 0$

$= -5$

$$G_x = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{900+25}$$

$$G = 30.41$$

$$G_x \begin{bmatrix} 30 & 35 \\ 50 & 55 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = 30 + 0 + 50 - 55$$

$$= -25$$

$$G_y = \begin{bmatrix} 30 & 35 \\ 50 & 55 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 0 + 35 - 50 + 0$$

$$= -15$$

$$G = \sqrt{625+225}$$

$$G = 29.15$$

$$G_x = \begin{bmatrix} 25 & 35 \\ 40 & 60 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = 25+0+0-60$$

$$= -35$$

$$G_y = \begin{bmatrix} 30 & 35 \\ 50 & 55 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 0+35-50+0$$

$$= -15$$

$$G = \sqrt{1225+225}$$

$$G = 38.07$$

$$G_x = \begin{bmatrix} 35 & 50 \\ 60 & 75 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = 35+0+0-75$$

$$= -40$$

$$\begin{bmatrix} 35 & 50 \\ 60 & 75 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 0 + 50 - 60 + 0$$

$$= -10$$

$$G = \sqrt{(40)^2 + 100}$$

$$G = 41.231$$

$$G_x = \begin{bmatrix} 50 & 55 \\ 75 & 85 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = 50+0+0-85$$
$$= -35$$

$$G_y = \begin{bmatrix} 50 & 55 \\ 75 & 85 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 0+55-75+0$$
$$= -20$$

$$G = \sqrt{1225+400}$$

$$G = 40.31$$

$$G_x = \begin{bmatrix} 40 & 60 \\ 50 & 70 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = 40+0+0-70$$
$$= -30$$

$$G_y = \begin{bmatrix} 40 & 60 \\ 50 & 70 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 0+60-50+0$$
$$= 10$$

$$G = \sqrt{900+100}$$

$$G = 31.62$$

$$G_x = \begin{bmatrix} 60 & 75 \\ 70 & 95 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = 60+0+0-95$$
$$= -35$$

$$G_y = \begin{bmatrix} 60 & 75 \\ 70 & 95 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 0+75-70+0$$
$$= 5$$

$$G = \sqrt{1225+25}$$

$$G = 35.35$$

$$G_x = \begin{bmatrix} 75 & 85 \\ 95 & 110 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = 75+0+0-110$$
$$= -35$$

$$G_y = \begin{bmatrix} 75 & 85 \\ 95 & 110 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 75(0)+85+(-95)+0$$
$$= -10$$

$$G = \sqrt{1225+100}$$

$$G = 36.40$$

Prewitt operator

* window size 3×3 matrix

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \quad G_y = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 25 & 40 & 55 \\ 45 & 65 & 85 \\ 70 & 95 & 120 \end{bmatrix}$$

$$G_x = \begin{bmatrix} 25 & 40 & 55 \\ 45 & 65 & 85 \\ 70 & 95 & 120 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} = -25+0+55 -45+0+ -70+85 +120$$

$$= 120$$

$$G_y = \begin{bmatrix} 25 & 40 & 55 \\ 45 & 65 & 85 \\ 70 & 95 & 120 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix} = 25+40+55+0+0+0 -70-95-120$$

$$= -165$$

$$G = \sqrt{(120)^2+(-165)^2}$$

$$G = \sqrt{14400+27225}$$

$$G = 204.02$$

approximate gradient

$$G = |G_x| + |G_y|$$

$$= 120+165$$

$$G = 285$$

Sobel Operator

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \qquad G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 10 & 90 \\ 15 & 15 & 95 \\ 20 & 20 & 100 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 10 & 90 \\ 15 & 15 & 95 \\ 20 & 20 & 100 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} = -10+0+90-30+0+190-20+0+100 = 320$$

$$\begin{bmatrix} 10 & 10 & 90 \\ 15 & 15 & 95 \\ 20 & 20 & 100 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} = 10+20+90+0+0+0+(-20)-40-100 = -40$$

$$G = \sqrt{102400+1600}$$

$$G = 322.49$$

approximation

$$G = |G_x| + |G_y|$$

$$= 320+40$$

$$G_{approx} = 360$$

Laplacian Operation

$$\begin{bmatrix} 20 & 30 & 40 & 50 \\ 30 & 45 & 60 & 75 \\ 45 & 60 & 80 & 95 \\ 60 & 80 & 100 & 120 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \rightarrow \text{Laplacian mask}$$

* Laplacian operation is 2nd order edge detection technique.
* It doesn't calculate gradient direction like sofell, prewitt, robert operation
* directly measures rate of intensity change
* If laplacian value 0 - low frequency else high frequency

$$L = 4C - (t+b+L+R)$$

t - top, b - bottom

L - left, R - Right

C - center

$$\begin{bmatrix} 20 & 30 & 40 \\ 30 & 45 & 60 \\ 45 & 60 & 80 \end{bmatrix} = 4(45) - (30+60+30+60)$$

$= 180 - 180$

$= 0$ (no edge)

$$\begin{bmatrix} 30 & 40 & 50 \\ 45 & 60 & 75 \\ 60 & 80 & 95 \end{bmatrix} = 4(60) - (40+80+45+75)$$

$= 240 - (240)$

$= 0$ (no edge)

$$\begin{bmatrix} 30 & 45 & 60 \\ 45 & 60 & 80 \\ 60 & 80 & 100 \end{bmatrix} = 4(60) - (45+45+80+80)$$

$= 240 - 250$

$= -10$ (dark edge)

$$\begin{bmatrix} 45 & 60 & 75 \\ 60 & 80 & 95 \\ 80 & 100 & 120 \end{bmatrix} = 4(80) - (60+60+95+0)$$

$= 320 - (120+195)$

$= 5$ (weak edge)

problem (20%) understanding	Method selection (20%)	Solution process (25%)	Accuracy of calculation (20%)	Intrepetation of results (15%)	Total
20	20	25	20	15	100